

Sam Tatum

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Education

Villanova University, Villanova, PA

August 2023 - May 2027

Bachelor of Science - Mechanical Engineering | GPA: 3.96 | Dean's List (all semesters attended)

Work Experience

Summer Intern, BCS Motion INC., Downingtown, PA

June 2024 - August 2025

- Designed, built, and wired an industrial electrical control panel and programmed an HMI interface to enable automated operation of a Kevlar braiding machine
- Collaborated on the design and integration of a custom automated sorting system using RoboUnits aluminum extrusion, an indexing belt conveyor, Asyrl vibration feeders, and a THL800 SCARA robot to detect, sort, and bag loose parts into kits
- Built an automated wheel-polishing system for 24-inch truck rims using Yaskawa servomotors and Unimotor ball-screw linear drives; developed a custom cleaning system to improve maintenance efficiency and increase system longevity
- Assisted in developing synchronized motion control sequences to prevent mechanical collisions and leveraged servomotor torque feedback for adaptive polishing pressure control
- Conducted an engineering decision matrix analysis to determine optimal suction cup end-effector sizing for robotic palletizing applications using a Yaskawa GP12 (6-axis industrial robot)
- Designed a custom mechanical end-stop and lift mechanism for an industrial conveyor system to improve product positioning accuracy and picking efficiency
- Managed and maintained bills of materials (BOMs) for multiple six-figure automation projects; tracked project inventory and reviewed 100+ engineering drawings to ensure design accuracy and manufacturing readiness

Engineering Experience

Volunteer, Fundación Un Cambio por la Vida, Santa Cruz (Galápagos), Ecuador

September - December 2025

- Co-organized and hosted a robotics showcase for 250+ attendees, increasing local receptivity to STEM education
- Created and taught robotics workshops using LEGO Mindstorms kits, instructing local high school students in fundamental programming and electromechanical systems concepts

Volunteer, Lifetime Wells, Ketu, Ghana

January - May 2025

- Assembled 3 hand-pump well heads to improve access to potable water in underserved communities
- Installed plunger assemblies with traveling valve and piston seals; deployed 296 feet of steel pump rod
- Conducted inspection, maintenance, and mapping of 200+ wells using ArcGIS to support infrastructure documentation and service planning

Volunteer, WindAid Institute, Trujillo, Peru

September - December 2024

- Manufactured fiberglass composite wind turbine blades and welded a structural steel tower for a small-scale wind energy system
- Fabricated and installed a wooden electrical enclosure integrating a DC-DC buck converter to regulate 12V turbine output to 5V power (usable electricity for lightbulbs and phone chargers)
- Collected and analyzed anemometer data from a 15-ft wind turbine installation in the Andes to assess site wind performance

Leadership

Resident Assistant, Villanova Residence Life

August 2024 - Present

- Fostered a safe and inclusive residential community while managing administrative responsibilities such as documentation, incident reporting, and coordination with professional staff to ensure student support and conflict mediation

Band Manager, Villanova Pep, Concert, and Jazz Band

March 2024 - Present

- Coordinated logistics, scheduling, communication, and equipment management for rehearsals and performances to ensure efficient event execution across multiple ensembles

Captain and Mechanical Lead, FTC First Tech Challenge Robotics Team 7244 Out of the Box

October 2017 - April 2023

- Led mechanical design, fabrication, and iteration of six World Championship-level competition robots under strict design constraints
- Integrated mechanical subsystems, including custom intake/outtake mechanisms, CNC-machined aluminum frames, and carbon fiber composite components
- Implemented PID-controlled drivetrain systems with functional odometry for precise motion control and autonomous navigation
- Led project management and engineering workflow, mentoring members through design reviews, prototyping, testing, and system innovation

Skills

Engineering & Technical: Finite Element Analysis (FEA); CAD (SolidWorks, Onshape); MATLAB; CNC machining (mill, lathe, router); arc welding; precision fabrication (bandsaw, drill press, grinders); industrial robots (Yaskawa GP12 6-axis, Shibaura THL800 SCARA); automated feeding systems (Asyrl Asycube); PLC/HMI programming (Asyrl Eye+ Studio, Weintek EasyBuilder Pro)

Activities: Black Belt (Tang Soo Do); Competitive Rowing (double & quad sculls); Musician (all brass instruments)

Languages: Spanish (working proficiency)